INTRODUCTION TO MICROBIOLOGY

DISCOVERY OF MICROORGANISMS

- Although Robert Hooke (1635-1703) published the first drawings of microorganisms in his book Micrographia, it is believed that the earliest observation of microbes was made by an Italian named Francesco Stelluti (1577-1652) using a microscope provided to him by Galileo
- Micrographia not only had detailed drawings of many microscopic things but also had information for building microscopes
- This piqued interest in many scientists of that generation who started building their own improved versions of the microscopes
- One among them was Antony van Leeuwenhoek (1632-1723) of Delft, the Netherlands who was a cloth merchant
- He designed microscopes that could provide magnifications up to 50 to 300 times
- Micrographia is important not only for its exquisite drawings but also for the information it provided on building microscopes.
- The microscopes designed by him were so powerful that they ultimately helped him observe and generate descriptions of both bacteria and Protists

LANDMARK DISCOVERIES RELEVANT TO THE FIELD OF MICROBIOLOGY

- A lookback at the history of microbiology reveals many important discoveries over years
- A few of the most import discoveries are listed below

Date	Microbiological History
1546	Fracastoro suggests that invisible organisms cause disease
1590–1608	Jansen develops first useful compound microscope
1676	Leeuwenhoek discovers "animalcules"
1688	Redi publishes work on spontaneous generation of maggots
1765–1776	Spallanzani attacks spontaneous generation
1786	Müller produces first classification of bacteria
1798	Jenner introduces cowpox vaccination for smallpox
1838–1839	Schwann and Schleiden, the Cell Theory
1835–1844	Bassi discovers that silkworm disease is caused by a fungus and proposes that many diseases are microbial in origin
1847–1850	Semmelweis shows that childbed fever is transmitted by physicians and introduces the use of antiseptics to prevent the disease
1849	Snow studies the epidemiology of a cholera epidemic in London

1857	Pasteur shows that lactic acid fermentation is due to a microorganism
1858	Virchow states that all cells come from cells
1861	Pasteur shows that microorganisms do not arise by spontaneous generation
1867	Lister publishes his work on antiseptic surgery
1869	Miescher discovers nucleic acids
1876–1877	Koch demonstrates that anthrax is caused by *Bacillus anthracis*
1880	Laveran discovers *Plasmodium,* the cause of malaria
1881	Koch cultures bacteria on gelatin
	Pasteur develops anthrax vaccine
1882	Koch discovers tubercle bacillus, *Mycobacterium tuberculosis*
1884	Koch's postulates first published
	Metchnikoff describes phagocytosis
	Autoclave developed
	Gram stain developed
1885	Pasteur develops rabies vaccine
	Escherich discovers *Escherichia coli,* a cause of diarrhea
1886	Fraenkel discovers *Streptococcus pneumoniae,* a cause of pneumonia

1887	Petri dish (plate) developed by Richard Petri
1887–1890	Winogradsky studies sulfur and nitrifying bacteria
1889	Beijerinck isolates root nodule bacteria
1890	Von Behring prepares antitoxins for diphtheria and tetanus
1892	Ivanowsky provides evidence for virus causation of tobacco mosaic disease
1894	Kitasato and Yersin discover *Yersinia pestis*, the cause of plague
1895	Bordet discovers complement
1896	Van Ermengem discovers *Clostridium botulinum*, the cause of botulism
1897	Buchner prepares extract of yeast that ferments
	Ross shows that malaria parasite is carried by the mosquito
1899	Beijerinck proves that a virus particle causes the tobacco mosaic disease
1900	Reed proves that yellow fever is transmitted by the mosquito
1902	Landsteiner discovers blood groups

1903	Wright and others discover antibodies in the blood of immunized animals
1905	Schaudinn and Hoffmann show *Treponema pallidum* causes syphilis
1906	Wassermann develops complement fixation test for syphilis
1909	Ricketts shows that Rocky Mountain spotted fever is transmitted by ticks and caused by a microbe (*Rickettsia rickettsii*)
1910	Ehrlich develops chemotherapeutic agent for syphilis
1911	Rous discovers a virus that causes cancer in chickens
1915–1917	D'Herelle and Twort discover bacterial viruses
1921	Fleming discovers lysozyme
1923	First edition of *Bergey's Manual*
1928	Griffith discovers bacterial transformation
1929	Fleming discovers penicillin
1931	Van Niel shows that photosynthetic bacteria use reduced compounds as electron donors without producing oxygen
1933	Ruska develops first transmission electron microscope
1935	Stanley crystallizes the tobacco mosaic virus Domagk discovers sulfa drugs
1937	Chatton divides living organisms into procaryotes and eucaryotes
1941	Beadle and Tatum, one-gene-one-enzyme hypothesis
1944	Avery shows that DNA carries information during transformation Waksman discovers streptomycin
1946	Lederberg and Tatum describe bacterial conjugation

1949	Enders, Weller, and Robbins grow poliovirus in human tissue cultures
1950	Lwoff induces lysogenic bacteriophages
1952	Hershey and Chase show that bacteriophages inject DNA into host cells
	Zinder and Lederberg discover generalized transduction
1953	Phase-contrast microscope developed
	Medawar discovers immune tolerance
	Watson and Crick propose the double helix structure for DNA
1955	Jacob and Wollman discover the F factor is a plasmid
	Jerne and Burnet propose the clonal selection theory
1959	Yalow develops the radioimmunoassay technique
1961	Jacob and Monod propose the operon model of gene regulation
1961–1966	Nirenberg, Khorana, and others elucidate the genetic code
1962	Porter proposes the basic structure for immunoglobulin G
	First quinolone antimicrobial (nalidixic acid) synthesized
1970	Discovery of restriction endonucleases by Arber and Smith
	Discovery of reverse transcriptase in retroviruses by Temin and Baltimore
1973	Ames develops a bacterial assay for the detection of mutagens
	Cohen, Boyer, Chang, and Helling use plasmid vectors to clone genes in bacteria
1975	Kohler and Milstein develop technique for the production of monoclonal antibodies
	Lyme disease discovered
1977	Recognition of archaea as a distinct microbial group by Woese and Fox

	Gilbert and Sanger develop techniques for DNA sequencing
1979	Insulin synthesized using recombinant DNA techniques
	Smallpox declared officially eliminated
1980	Development of the scanning tunneling microscope
1982	Recombinant hepatitis B vaccine developed
1982–1983	Discovery of catalytic RNA by Cech and Altman
1983–1984	The human immunodeficiency virus isolated and identified by Gallo and Montagnier
	The polymerase chain reaction developed by Mullis
1986	First vaccine (hepatitis B vaccine) produced by genetic engineering approved for human use
1990	First human gene-therapy testing begun
1992	First human trials of antisense therapy
1995	Chickenpox vaccine approved for U.S. use
	Haemophilus influenzae genome sequenced
1996	*Methanococcus jannaschii* genome sequenced
	Yeast genome sequenced
1997	Discovery of *Thiomargarita namibiensis,* the largest known bacterium
	Escherichia coli genome sequenced
2000	Discovery that *Vibrio cholerae* has two separate chromosomes

2001 Human Genome Sequencing

2004 Reconstruction of whole genomes of unculturable microorganisms

2005 First comprehensive metaproteomics study of a microbial community

2007 The first phase of the human microbiome started

2008 Short reads technology

2010 Bovine rumen microbiome

2011 Earth Microbiome project foundation

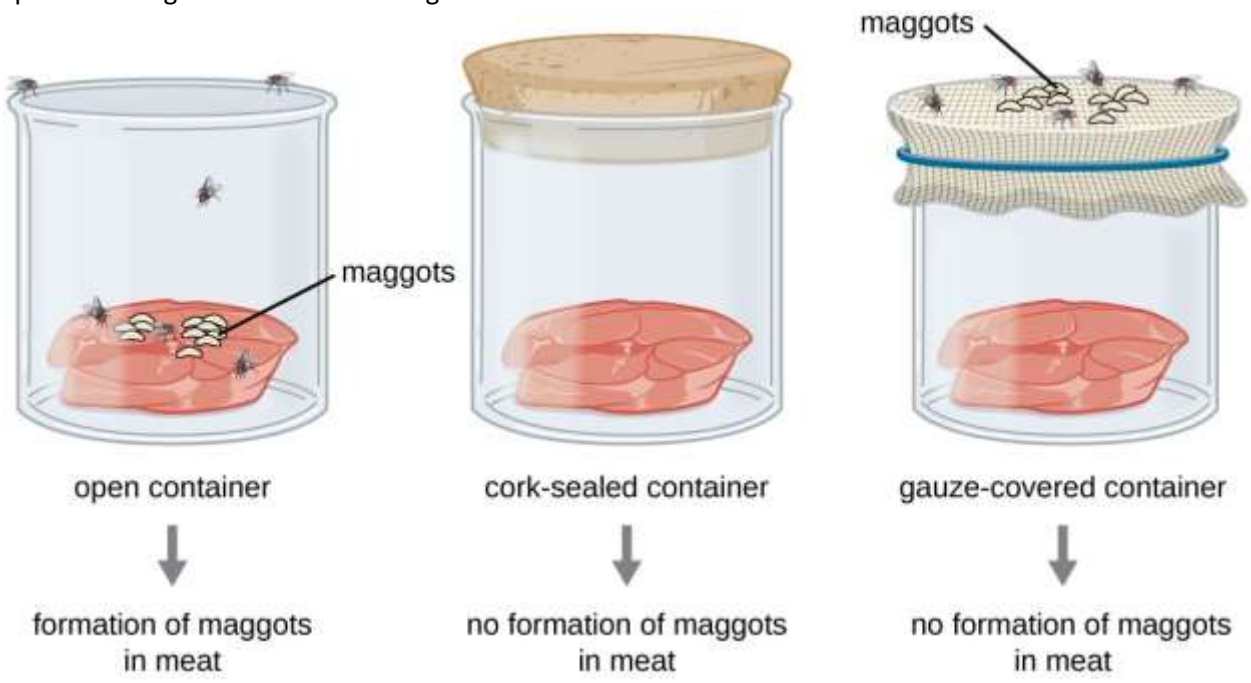
2012 First phase of Human Microbiome project started

2013 First Human Microbiome transplantation

2017 Multi-Omic analysis as forensic tool

SPONTANEOUS GENERATION OF MICROORGANISMS

- In the beginning when people lacked enough information about the origin of living organisms, they believed that living organisms were developed from nonliving matter
- This belief or hypothesis that living organisms were derived from nonliving materials is called the spontaneous generation of microorganisms
- The spontaneous generation theory was disproved by an Italian physician named Francesco Redi (1626-1697), who carried out a series of experiments on decaying meat
- Before disproving the spontaneous theory, people believed that maggots (larva of flies that develop from eggs laid by flies) were generated spontaneously on the decaying meat
- To disprove this, he took the meat in three containers, he sealed one container with cork, the other with gauge that will not allow flies and the third one was left uncovered
- He showed that in the meat sealed with cork no maggots were formed there by disproving the spontaneous generation of microorganisms



- Similar experiments by others helped discredit the theory for larger organisms
- Louis Pasteur (1822-1895) once and for all settled the matter of spontaneous generation using his Swan-neck flasks experiment

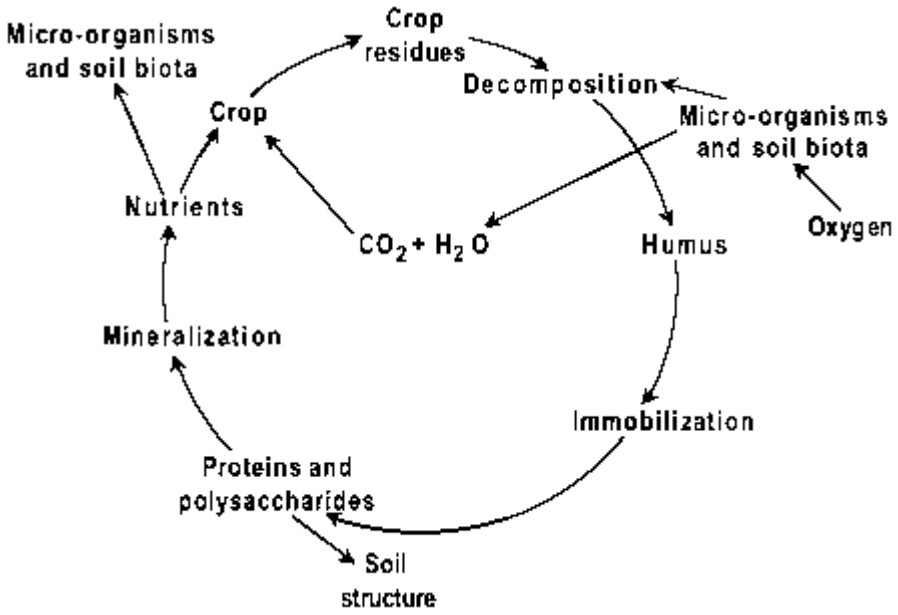
Methodology	Control Results	Experimental Results
		
Broth in flask is boiled to kill pre-existing micro-organisms (create a sterile environment)	As broth cools, condensing water collects, sealing mouth of flask (no growth will occur)	If neck is broken, outside air can carry micro-organisms into broth (contamination)

Conclusion

Cells can ***only*** arise from pre-existing cells

ROLE OF MICROORGANISMS IN THE TRANSFORMATION OF ORGANIC MATTER

- Microorganisms play an important role not only in the transformation of organic compounds but also in the transformation of radionuclides and toxic metals present in the waste.
- The organic material such as leaves, roots, dead organisms, etc. shed into the layers of soil are broken down to a reusable organic matter by microorganisms present in the soil
- These microorganisms include both bacteria and fungi that have the ability to produce various kinds of enzymes necessary for degrading these organic materials
- the nature of the organic matter that is generated depends on various factors like age, mode of transformation, and existing environment
- when organic matter is finally broken down a stable substance that resists further decomposition called the humus is produced
- This process is called humification
- Soil organisms, including micro-organisms, use soil organic matter as food.
- In the process of breaking down any organic matter, any excess nutrients (N, P and S) are released into the soil in forms that plants can use.
- This release process is called mineralization.
- So, the transformation of organic matter by microorganisms results in humification and mineralization which can then be used by plants and other organisms to continue the cycle of life



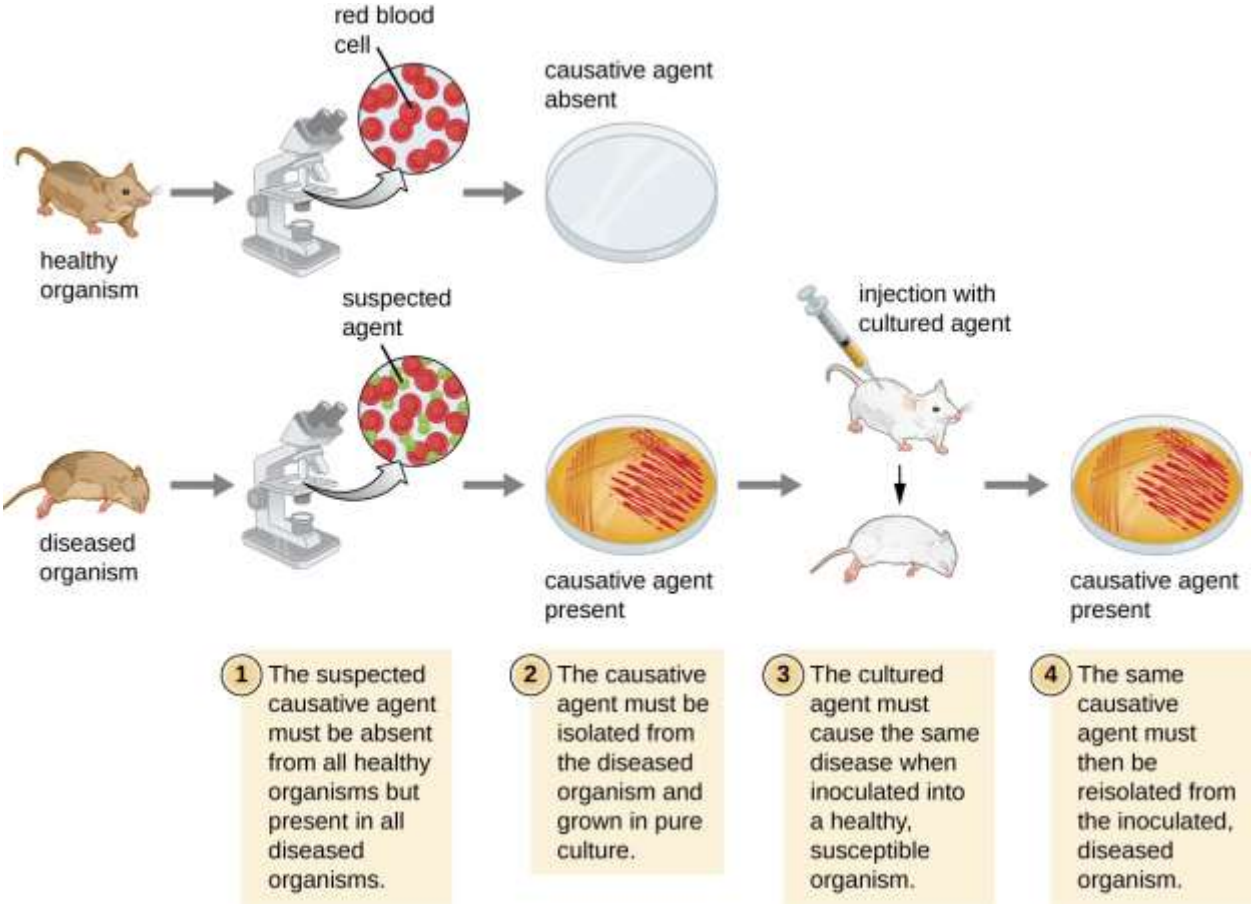
MICROORGANISMS AND DISEASE

- Fracastoro and a few others for the first time suggested that invisible organisms are responsible for the diseases
- This belief by many scientists of that generation eventually lead to the development of germ theory of disease
- Agostino Bassi (1773-1856) in 1835 showed that a silkworm disease was caused due to fungal infection.
- M. J. Berkeley (1803-1889) proved that the great potato blight of Ireland was caused by a water mold
- Heinrich de Bary a German surgeon and botanist in 1853 (1831-1888) showed that smut and rust fungi caused cereal crop diseases.
- Joseph lister provided indirect evidence to support germ theory. He developed a system of antiseptic surgery i.e. he prevented sepsis which results from growth of microorganisms in the wounds by ensuring that the instruments were heat sterilized (heat kills microorganisms) and employed phenol on surgical dressings and sometimes sprayed over the surgical area
- This not only led to many successful surgeries in the years to follow but also led to providing an indirect proof that destroying microorganisms by heating and phenol treatment prevented infections and subsequent sepsis caused by microorganisms

KOCH'S POSTULATES

- It was Robert Koch who first demonstrated that bacteria are responsible for the disease

- His experiments with Bacillus anthracis that causes anthrax ultimately proved that microorganisms are responsible for the diseases
- Based on his experiments he proposed some postulates which are commonly referred to as Koch's postulates
- These postulates help in establishing a relation between the disease and the causative organism of that disease.
- An organism is said to be responsible for a particular disease only when the postulates are fully met
- The four Koch's postulates are shown in the image below



Important notes: The most important aspect in this chapter is the Koch's postulates. The rest of the information is important for understanding the history and development of the subject better but not so much in the exam point of view